

The Extension Problem for Orthomodular Observation Algebras

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1 Background

All measurement is finite. An experimenter performs finitely many trials, records finitely many digits, distinguishes finitely many outcomes. Yet the mathematical frameworks we use to organise empirical knowledge — probability theory, quantum mechanics, statistical mechanics — rest on infinite structures: σ -algebras, Hilbert spaces, continuous symmetry groups. The passage from finite observation to infinite structure is not automatic; it requires a commitment that no finite body of evidence can compel.

In probability theory, the locus of this commitment is *countable additivity*. Given a Boolean algebra $\mathcal{E}$ of events and a finitely additive charge $\ell : \mathcal{E} \rightarrow [0, 1]$ with $\ell(\Omega) = 1$, the condition

$$E_n \downarrow \emptyset \implies \ell(E_n) \rightarrow 0 \quad (1)$$

— that mass cannot persist along a vanishing sequence of events — is not derivable from any finite collection of observations. De Finetti [7] argued that only finite additivity is empirically grounded; Kolmogorov [15] adopted σ -additivity as an axiom and built modern probability on it. The question of *when* and *why* the passage from one to the other is justified has never been fully resolved.

In quantum theory, the same tension appears in a different guise. The algebra of propositions about a quantum system is not Boolean: incompatible observables cannot be simultaneously determined. The closed subspaces of a Hilbert space H form an orthomodular lattice $L(H)$ — a structure in which complementation is well-defined but distribution of meet over join may fail:

$$a \wedge (b \vee c) \neq (a \wedge b) \vee (a \wedge c) \quad \text{in general.} \quad (2)$$

A *state* on such a lattice is a function $s : L(H) \rightarrow [0, 1]$ satisfying

$$s(\mathbf{1}) = 1, \quad s(a \vee b) = s(a) + s(b) \quad \text{whenever } a \perp b. \quad (3)$$

The non-distributivity means that s is additive only on orthogonal pairs, not on arbitrary finite unions. The classical machinery for extending finitely additive assignments to σ -additive ones — Carathéodory's outer-measure construction, which requires a Boolean algebra of sets — does not apply.

For the specific lattice $L(H)$, this difficulty is resolved by a remarkable rigidity. Gleason's theorem [12] shows that, when $\dim H \geq 3$, every state has the form

$$s(P) = \text{tr}(\rho P) \quad (4)$$

for a unique density operator ρ , and is therefore automatically σ -additive. The geometry of Hilbert space is so constraining that the problem of the infinite simply does not arise — states have no room to misbehave.

But Hilbert space is very special. Orthomodular lattices arise in many other settings — the projection lattices of operator algebras, the logics of quantum systems with superselection rules, and the abstract lattices studied in quantum logic — and for these, no analogue of Gleason’s rigidity is known. The question that probability theory and quantum theory share — when does finite coherence force countable behaviour? — remains open in this broader setting, and is the subject of the present note.

The programme “Structure from Observation” [17] approaches this question from a particular direction: rather than beginning with a pre-given space Ω of outcomes and asking which σ -algebras it supports, it begins with the *observations themselves* — a directed system of event algebras $\mathcal{E}_i$ with compatible charges ℓ_i — and asks what the directed structure alone determines.

Paper I answers this for Boolean event algebras. A compatible family of finitely additive charges extends to a σ -additive probability measure if and only if collective exhaustion (CE) holds — condition (1) applied uniformly across the directed system. Two constructions realise the extension:

- **Carathéodory:** assumes CE; produces a measure on $(\Omega, \sigma(\mathcal{C}))$ directly.
- **Stone:** assumes nothing; produces a Baire measure on the Stone space $\text{St}(\mathcal{C})$ unconditionally.

Under discriminability, the two agree. Three questions orient the broader programme:

- (1) What must be added to observation to get structure?
- (2) Reconstruction without points: what structure is already present in the relational description — contexts, directed refinement — before point-spaces are assumed?
- (3) Local-to-global extension: what conditions on the *site* control whether local data extend globally?

Paper I resolves (2) and (3) for Boolean algebras. The present note asks whether any of this survives when the event algebra is orthomodular but not distributive.

Epperson and Zafiris [10] advocate a *relational realist* ontology in which structure is constituted by relations among contexts, not by points in a pre-given space. An orthomodular lattice A is the natural algebraic expression of this: its elements are propositions, its partial order is entailment, and its non-distributivity reflects the impossibility of assigning simultaneous truth values to all propositions. The dual space $S_0(A)$ of McDonald–Bimbó [18] is built from *filters* — consistent collections of propositions — not from points of a pre-existing space. This is question (2) applied to non-Boolean algebras: what structure does the relational description already contain?

In the Boolean case, one might hope that the passage from finite to countable additivity could be understood as a *sheaf condition* — local states gluing to a global σ -additive measure. Biesel [1] shows this hope is disappointed:

- Finitely additive charges form a sheaf for finite covers.
- σ -additive measures form a sheaf for countable covers.
- The step between is the *definition* of σ -additivity, not a structural theorem.

This is the boundary at which the present problem begins.

Paper II [16] introduces three positions on the relationship between observation and reality, each a constraint on where the dual-space measure lives:

Empirical adequacy (EA)

The dual-space measure agrees with all observations.

Probabilistic realism (PR)

The measure descends to a realisation space.

Value-definite realism (VDR)

A global two-valued homomorphism exists.

These are nested: $VDR \Rightarrow PR \Rightarrow EA$. For Boolean algebras, all three are commensurable — every EA-theory admits a VDR-completion. For OMLs with $\dim \geq 3$, Gleason’s theorem (4) rescues PR, but Kochen–Specker [14] blocks VDR. The extension problem sits at the $EA \rightarrow PR$ transition: can we pass from a state on the algebra (EA) to a σ -additive measure on the dual space (PR)? If so, under what conditions — and is descent then a free choice (as in the Boolean case) or a structural consequence of the algebra and state together?

Van Fraassen’s distinction [25] between empirical adequacy and structural commitment, which Paper I makes precise via the Stone construction, thus acquires a sharper form in the OML setting: the Stone route is no longer available, the Carathéodory route is blocked by non-distributivity, and the passage from coherent local data to global probability — question (3) — requires genuinely new ideas.

2 The problem

Let A be an orthomodular lattice (OML) and let $s : A \rightarrow [0, 1]$ be a state (3). The McDonald–Bimbó duality [18] associates to A a compact topological space

$$S_0(A) = (F(A), \subseteq, \perp_A, P(A), T(S)),$$

where $F(A)$ is the set of filters on A and $P(A) \subseteq F(A)$ is the set of prime filters. For each $a \in A$, set

$$h(a) = \{x \in F(A) : a \in x\}.$$

The map h is an OML isomorphism from A onto the $\perp$ -stable clopen sets $\text{CO}(S_0(A))^\dagger$. Define a set function μ on these clopens by

$$\mu(h(a)) = s(a). \tag{5}$$

Question 2.1 (Extension). Under what conditions on the OML A does μ extend to a σ -additive measure on the Baire (or Borel) σ -algebra of $S_0(A)$?

This is the direct OML analogue of the Stone extension theorem established for Boolean algebras in [17].

In the Boolean case, the extension is automatic. The clopens $\text{Clop}(\text{St}(\mathcal{C}))$ form a Boolean algebra, the state s restricts to a finitely additive charge on this algebra, and the Carathéodory extension theorem (or Choksi’s projective-limit argument [6]) produces the σ -additive extension. Compactness of the Stone space gives σ -subadditivity for free.

In the OML case each of these steps breaks:

- (i) $\text{CO}(S_0(A))^\dagger$ is an OML, *not* a Boolean algebra.
- (ii) The state s is additive on orthogonal pairs only, not on arbitrary finite unions.
- (iii) **Carathéodory does not apply** — there is no Boolean algebra of sets on which to run the outer-measure construction.
- (iv) Compactness holds (McDonald–Bimbó, Lemma 3.5 [18]), but the σ -subadditivity argument requires orthogonality structure that the topology alone does not provide.

The gap is sharp: “finitely additive on a compact Boolean algebra of clopens” automatically extends; “orthogonally additive on a compact OML of clopen $\perp$ -stable sets” does *not*.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason’s theorem (4) resolves the problem completely: every state is induced by a density operator, and the resulting measure is σ -additive on $L(H)$. The extension to $S_0(L(H))$ follows. For general OMLs, no Gleason-type representation is available. Question 2.1 asks for the structural conditions that substitute for the Hilbert-space geometry.

Feature	Boolean (Paper I)	OML (this problem)
Dual space	$\text{St}(A)$ (ultrafilters)	$S_0(A)$ (all filters)
Distinguished subset	$\text{pure}(\Omega)$, not canonical	$P(A)$, canonical
Clopes	Boolean algebra	OML (non-distributive)
Additivity of s	full (finite)	orthogonal pairs only
Extension	automatic (Carathéodory)	requires Gleason-type result
Realisation constrained?	No	Yes (Kochen–Specker [14])

Table 1: Structural comparison of the extension problem.

Two axes: extension and descent. The passage from a state on A to a measure on $S_0(A)$ splits into two axes that the Boolean case fuses but the OML case separates. Since $S_0(A)$ is a Stone space, its *full* clopen algebra $\text{CO}(S_0(A))$ is Boolean, whereas the state lives only on the $\perp$ -stable clopens $\text{CO}(S_0(A))^\dagger$ — the OML, the image of h . These share only meet; join, complement, and bottom differ.

Extension Extend μ from the $\perp$ -stable clopens to a finitely additive charge on the *full* Boolean algebra of clopens — the classical Boolean extension problem [13]. Since h preserves meets, $a \wedge b = 0$ forces $h(a) \cap h(b) = h(0)$, so meet-zero elements of A map to *disjoint* clopens. A charge must therefore satisfy, for every pairwise-meet-zero family $\{a_i\}$, $\sum_i s(a_i) \leq 1$ — a Pitowsky correlation polytope / Bell–Boole inequality system [20]. σ -additivity of the state does not help here.

Compactness

Once extension succeeds, the charge extends to a σ -additive Borel measure automatically. Unconditional.

In the Boolean case (Paper I), the extension axis is *free*: finite additivity on the clopen Boolean algebra plus compactness gives the Stone measure unconditionally — “the first arrow is free.” In the OML case, **the extension axis is no longer free.**

The extension obstruction is meet-zero versus orthogonal. Three strengthening conditions on s must be distinguished (Pták–Pulmannová [23], Def. 2):

- (1) *State*: additive on *orthogonal* pairs ($a \leq b^\perp$).
- (2) *Valuation*: additive on *meet-zero* pairs ($a \wedge b = 0$) — a binary condition, strictly stronger, since in a non-distributive OML meet-zero is coarser than orthogonality.
- (3) *Extends to a Boolean charge*: the n -ary condition $\sum_i s(a_i) \leq 1$ for every pairwise-meet-zero family — strictly stronger again.

The extension axis is condition (3). It can fail for *every* state. On the smallest non-distributive OML MO_3 (three 2×2 blocks pasted at 0 and 1), all six atoms are pairwise meet-zero, so orthoadditivity forces the three complementary pairs to sum to 3 while a charge demands ≤ 1 : no state extends. The maximally mixed state $s \equiv \frac{1}{2}$ is a valuation (it satisfies the binary condition (2)) yet already fails the n -ary condition (3) at the triple $\{p, q, r\}$, where $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So the obstruction is the gap between meet-zero and orthogonality, not σ -additivity and not non-contextuality.

Status of the extension axis. For *concrete* (set-representable) OMLs the lattice meet is set intersection, so meet-zero coincides with orthogonality, and the extension axis reduces to the classical marginal / Pitowsky non-contextuality problem. For *non-concrete* OMLs (MO_3 , $L(H)$) it is strictly stronger and can fail universally. In the *finite* case it is a decidable linear program, not a structurally resistant frontier. What remains open is the *infinite* case — $L(H)$ and the actual σ -additive measure on $S_0(A)$ — to which the finite counterexample does not speak.

Descent. σ -additivity of the state governs descent, not extension. Even once μ extends to a measure $\hat{\mu}$ on $S_0(A)$, the *descent* question — whether $\hat{\mu}$ concentrates on the “physical” points — changes character:

Boolean	Does $\hat{\mu}$ concentrate on $\text{pure}(\Omega)$? If and only if the charges are σ -additive. The choice of realisation space Ω is unconstrained.
OML	Does $\hat{\mu}$ concentrate on $P(A)$? Here $P(A)$ is part of the structure, not a free choice. The Kochen–Specker obstruction [14] prevents simultaneous realisation of all filters; the Bub–Clifton theorem [2] shows that the state determines a maximal Boolean subalgebra of definite values.

In the OML setting, the algebra and state *together* constrain which filters are “realised.” Descent is not a geometric commitment but a structural consequence.

Remark 2.2 (Decomposition lives on the descent axis). De Simone–Navara [8] decompose OMP states as $s = s_\sigma + s_{\text{wpfa}}$ (a σ -additive part and a weakly purely finitely additive part), the orthomodular analogue of Yosida–Hewitt. By the axis split above, the wpfa component governs the *descent* axis — it is the analogue of the purely finitely additive part ℓ_p in the Boolean case — and *not* the extension axis. A natural-looking conjecture, “ s extends iff $s_{\text{wpfa}} = 0$,” is therefore mis-targeted: vanishing of s_{wpfa} concerns whether the (already σ -additive) state’s measure concentrates on $P(A)$, while extension is blocked by non-distributivity regardless of the wpfa component. The decomposition theory and the extension problem live on different axes.

A caveat compounds this. The McDonald–Bimbó duality [18] is *finitary*: the identity $h(\bigvee_n a_n) = (\bigcup_n h(a_n))^{\perp\perp}$ holds for finite joins but fails in general for countable joins, since finitary OML homomorphisms need not preserve infinite suprema. Making even the descent argument rigorous — the analogue of Paper I’s “ σ -additivity $\Leftrightarrow$ concentration” — requires σ -completeness and continuity hypotheses the 2023 duality does not supply, and is itself open.

3 What is known

In the Boolean case, the Kelley–Vladimirov–Pták characterisation (Fremlin, Theorem 391D [11]) gives a complete algebraic answer:

Theorem 3.1 (KVP). *A Boolean algebra B carries a strictly positive σ -additive measure if and only if B is*

- (a) *Dedekind σ -complete*,
- (b) *weakly (σ, ∞) -distributive*, and
- (c) *chargeable (carries a strictly positive finitely additive measure)*.

Each condition is algebraic and non-trivially constraining. For OMLs, the analogues fail or degenerate:

- (a) Dedekind σ -completeness makes sense but is very strong.
- (b) Weak (σ, ∞) -distributivity has no clean OML analogue — the definition relies on distributivity of $\bigwedge$ over $\bigvee$, which may fail orthomodularly.
- (c) Chargeability has no known structural characterisation on OMLs; the existence of a strictly positive finitely additive measure cannot be read off from the lattice structure.

The positive results that do exist for OMLs all require algebraic structure rich enough to approximate Hilbert-space geometry:

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| Gleason [12] | $A = L(H)$, $\dim H \geq 3$. Every state has the form (4); σ -additivity is automatic. |
| Bunce–Wright [3] | A the projection lattice of a JBW-algebra. σ -additivity via operator-algebraic structure. |
| Chetcuti–Dvurečenskij [5] | Lattice effect algebras with completeness conditions close to von Neumann algebras. |

The obstructive results show that the Boolean extension machinery cannot be transplanted:

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|-----------------------------|---|
| Pták–Pulmannová [22] | If every unital subadditive measure on an orthomodular poset is a state, the poset must be Boolean. This is the structural reason no KVP analogue exists for OMLs: conditions strong enough to force σ -additivity collapse the lattice back to a Boolean algebra. |
| Navara [19] | Regularity does <i>not</i> force σ -additivity on general orthomodular posets. The Béaver–Cook result is special to von Neumann algebras. |
| Pták [21] | Exotic OML state spaces exist; the obstruction is combinatorial (Greechie pasting), not from the centre. |

Recent incremental work confirms the field is active but no one has found the right condition: Voráček–Pták [26] study signed measures on orthomodular posets; Burešová–Pták [4] investigate variations on regularity conditions; De Simone–Navara study Yosida–Hewitt-type decompositions for orthomodular posets.

For the σ -additivity / descent side, the gap between “too weak” (allows non- σ -additive states) and “too strong” (collapses to Boolean) appears structurally robust — this is the part where further reading has not produced the answer. (This robustness is about the descent axis; the extension axis is classical, per §2.)

All known exotic OML state spaces are built by **Greechie pasting** [27, 24, 21]. Progress on the descent residue likely requires either:

- (a) a new pasting technique that controls σ -additivity, or
- (b) an approach that bypasses the state space entirely (e.g., categorical or topos-theoretic [9]).

4 Assessment

Question 2.1, once split into its two axes, is not a single open problem but a settled axis and an open one.

Extension axis: largely settled, and partly degenerate. This is the classical Boolean extension problem [13] / Pitowsky polytope feasibility [20]. Because h preserves meets, it is governed by the gap between meet-zero and orthogonality, which in a non-distributive OML is non-empty. In the finite case it is a decidable linear program; it can fail for *every* state (MO_3). This is not a structurally resistant frontier and does not require a new idea. The earlier framing of this axis as “the Pták–Pulmannová frontier needing a new idea” was mistaken — Pták–Pulmannová [23] concerns the *supply* of valuations (a unital set forces Boolean), not whether a given state extends, and extension is in any case strictly stronger than the valuation condition.

Descent axis: genuinely open. Whether a measure on $S_0(A)$ concentrates on the physical points $P(A)$, and the analogue of Paper I’s “ σ -additivity $\Leftrightarrow$ concentration,” cannot even be stated rigorously without σ -completeness and continuity hypotheses, because the McDonald–Bimbó duality is finitary (the countable-join identity fails). This is the live residue.

The infinite extension case. The finite counterexample does not speak to $L(H)$ or to the actual σ -additive measure on $S_0(A)$ for an infinite OML. Whether the meet-zero/orthogonal gap produces the same universal failure infinitely, or whether σ -completeness changes the picture, is open.

Novelty. The formulation via McDonald–Bimbó duality [18] appears to be new; the identification of the extension axis with classical Horn–Tarski / Pitowsky feasibility de-mystifies it. For $L(H)$, Gleason resolves both axes at once.

Status. Extension axis: settled (finite), open (infinite). Descent axis: open, contingent on a rigorous finitary-to- σ bridge. Not a current active lead.

References

- [1] Owen Biesel. Sheaves of probability. arXiv preprint arXiv:2401.01968, 2024.
- [2] Jeffrey Bub and Rob Clifton. A uniqueness theorem for ‘no collapse’ interpretations of quantum mechanics. *Studies in History and Philosophy of Modern Physics*, 27(2):181–219, 1996.
- [3] Leslie J. Bunce and J. D. Maitland Wright. The Mackey–Gleason problem. *Bulletin of the American Mathematical Society*, 26(2):288–293, 1992.
- [4] Klára Burešová and Pavel Pták. Variations on regularity conditions for orthomodular posets. 2023. Preprint.
- [5] Emmanuel Chetcuti and Anatolij Dvurečenskij. On the Gleason theorem for effect algebras. *Mathematica Slovaca*, 55(3):297–308, 2005.
- [6] J. R. Choksi. Inverse limits of measure spaces. *Proceedings of the London Mathematical Society*, 8(3):321–342, 1958.
- [7] Bruno de Finetti. *Theory of Probability*, volume 1. Wiley, 1974.
- [8] Anna De Simone and Mirko Navara. Yosida–Hewitt and Lebesgue decompositions of states on orthomodular posets. *Journal of Mathematical Analysis and Applications*, 255(1):74–104, 2001.

- [9] Andreas Döring and Chris J. Isham. A topos foundation for theories of physics: I. Formal languages for physics. *Journal of Mathematical Physics*, 49(5):053515, 2008.
- [10] Michael Epperson and Elias Zafiris. *Foundations of Relational Realism: A Topological Approach to Quantum Mechanics and the Philosophy of Nature*. Lexington Books, Lanham, MD, 2013.
- [11] D. H. Fremlin. *Measure Theory, Vol. 3: Measure Algebras*. Torres Fremlin, Colchester, 2004.
- [12] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.
- [13] Alfred Horn and Alfred Tarski. Measures in Boolean algebras. *Transactions of the American Mathematical Society*, 64(3):467–497, 1948.
- [14] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.
- [15] A. N. Kolmogorov. *Foundations of the Theory of Probability*. Chelsea, 1933.
- [16] Patrick S. Mahon. Distributivity and the commensurability of empirical adequacy and realism. Preprint, 2026.
- [17] Patrick S. Mahon. When does observational coherence determine probability? Preprint, 2026.
- [18] J. McDonald and Katalin Bimbó. Duality for orthomodular lattices. 2023. Preprint.
- [19] Mirko Navara. When is the integral on quantum probability spaces additive? *Bulletin of the Polish Academy of Sciences — Mathematics*, 40(1):29–33, 1992.
- [20] Itamar Pitowsky. *Quantum Probability — Quantum Logic*, volume 321 of *Lecture Notes in Physics*. Springer, Berlin, 1989.
- [21] Pavel Pták. Exotic logics. *Colloquium Mathematicum*, 54(1):1–7, 1987.
- [22] Pavel Pták and Sylvia Pulmannová. *Orthomodular Structures as Quantum Logics*, volume 44 of *Fundamental Theories of Physics*. Kluwer, Dordrecht, 1991.
- [23] Pavel Pták and Sylvia Pulmannová. A measure-theoretic characterization of Boolean algebras among orthomodular lattices. *Commentationes Mathematicae Universitatis Carolinae*, 35(1):205–208, 1994.
- [24] Frederic W. Shultz. A characterization of state spaces of orthomodular lattices. *Journal of Combinatorial Theory, Series A*, 17(3):317–328, 1974.
- [25] Bas C. van Fraassen. *The Scientific Image*. Oxford University Press, 1980.
- [26] Václav Voráček and Pavel Pták. Signed measures on orthomodular posets. 2023. Preprint.
- [27] Alexander Wilce. Test spaces. In Kurt Engesser, Dov M. Gabbay, and Daniel Lehmann, editors, *Handbook of Quantum Logic and Quantum Structures: Quantum Logic*, pages 443–549. Elsevier, 2009.